PROBLEM 35

String with Two Particles on a Cylinder

Original statement · Problems.pdf, p. 23

Multi-Particle Cylinder Contact

Two heavy particles P and Q of mass m each are connected by an inextensible massless string of constant length L. Part of this string rests on top of a cylinder of radius R and center at O. The string can slide over the cylinder without friction, and remains tense at all times. Furthermore, P and Q are also connected by a spring of zero natural length and stiffness k. We define the inertial reference frame S_0. We denote as A and B the points of tangency of the string with the cylinder.

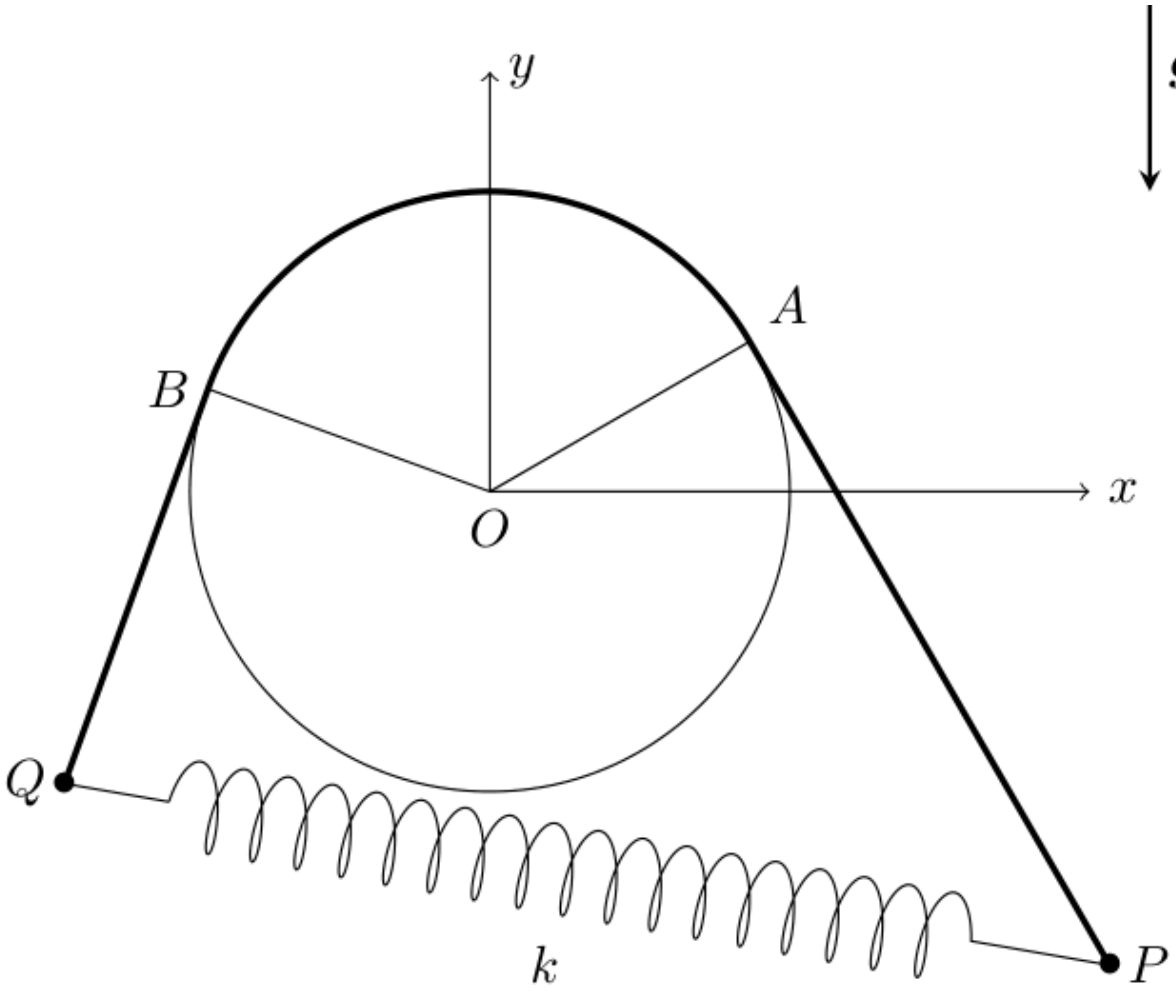


Figure: String over the cylinder of radius R centred at O with tangency points A and B, particles P and Q joined by the spring k (Problems.pdf, p. 23).

(a) How many degrees of freedom does this system have? Define your own generalized coordinates to parametrize the configuration of the system. You will use them in the rest of the problem.

(b) Write down the general expression of the position, velocity, and acceleration vectors of P and Q relative to the S_0 reference frame as a function of your chosen coordinates and their time derivatives.

(c) Write down the expression of all forces acting on P and Q.

(d) Discuss whether the mechanical energy of P is conserved; if so, write down the expression of the corresponding potential energy.

(e) Consider now that we restrict the motion of the system to symmetric configurations about the Oy axis only. What is the number of degrees of freedom now? Find the mathematical relations that exist between your previously-defined coordinates in this case.

(f) Find the approximate value of your coordinates when the system is in equilibrium and the configuration is symmetric about the Oy axis. To do so, assume that the deviation from the vertical of the AP and BQ segments is small, and linearize the equations of motion. Find the angular frequency of the oscillations around this equilibrium configuration.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two particles P and Q of mass m connected by an inextensible massless string of length L that passes over a smooth horizontal cylinder of radius R.
- In addition, P and Q are connected by a spring of zero natural length and stiffness k.
- Tangency points with the cylinder are A (left) and B (right).
- **CDOFs:** System in plane: 2 degrees of freedom.
- **Parameters:** $m, R, L, k, g > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Frame S_0 centered at O.
- Cylinder tangency points: A at angle θ, B at angle ψ.
- Length of string: $L = \xi_P + R(\psi - \theta) + \xi_Q$.
- Connecting vector $\mathbf{PQ} = \mathbf{r}_Q - \mathbf{r}_P$.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a)-(c): Forces and Spring Interaction

Spring force on P and Q:

$$\mathbf{F}_{S,P} = -\mathbf{F}_{S,Q} = k\mathbf{PQ} = k(\mathbf{r}_Q - \mathbf{r}_P)$$

Expressing $\mathbf{r}_P$ and $\mathbf{r}_Q$ via tangency points:

$$\mathbf{F}_{S,P} = kR(\mathbf{e}_B - \mathbf{e}_A) + k[L - \xi - R(\psi - \theta)]\mathbf{e}_\psi + k\xi\mathbf{e}_\theta$$

Compare with the official solution key (Problems.pdf, Solution keys section).

Part (d)-(f): Symmetric Configurations and Small-Angle Oscillations

For symmetric configurations about Oy: $\psi = \pi - \theta$ and $\xi_P = \xi_Q$.

The system reduces to **1 degree of freedom**.

Linearizing the equations of motion for small angular deviations $\delta\theta$ around the symmetric equilibrium yields a simple harmonic oscillator with natural frequency:

$$\omega_n = \sqrt{\frac{g}{R} + \frac{2k}{m}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- The spring acts in parallel with gravity, stiffening the restoring torque and raising the natural frequency above the gravitational pendulum value $\sqrt{g/R}$.

Source: Problems.pdf (Page 23)